

1. ABSTRACT

This project explores the correlation between ambient light intensity and human emotional response by utilizing a compact, sensor-driven embedded system based on a Raspberry Pi. Emotional changes are detected using the BioAmp EXG Pill, a biomedical signal acquisition module capable of capturing electrocardiogram (ECG) and electroencephalogram (EEG) signals from the human body. These physiological signals, which vary according to the individual's emotional state, are captured in real-time and analyzed in conjunction with ambient light intensity data acquired from a light-dependent resistor (LDR). As the LDR produces analog voltage corresponding to the surrounding light level, an analog-to-digital converter (ADS1115) is employed to digitize the LDR output for processing by the Raspberry Pi.

The project architecture includes the integration of the EXG module and LDR with the Raspberry Pi through carefully constructed electronic interfacing. The EXG module is connected via GPIO pins, capturing the biopotential signals, while the ADS1115 ADC uses the SPI interface to digitize LDR sensor data. The system collects simultaneous readings from both input sources to establish a time-correlated dataset of emotional response and lighting conditions. A key feature of this system is its ability to visualize the collected data in real time using Python and the Matplotlib library. Graphical plots show both EEG/ECG waveform changes and fluctuations in light intensity over time, helping to infer potential emotional triggers or patterns in relation to lighting conditions.

Furthermore, the project incorporates a programmable LED lighting system controlled through PWM (Pulse Width Modulation) signals generated by the Raspberry Pi. This allows dynamic changes in ambient lighting levels to test and observe the subject's emotional response to varying light intensities and colours. As an optional enhancement, machine learning algorithms can be integrated to classify emotional states from the EXG signal patterns, opening the door for adaptive lighting systems that respond to user emotions.

The primary objectives of this project are to:

- (1) demonstrate a functional embedded system capable of bio-signal monitoring and environmental sensing,
- (2) investigate how environmental lighting conditions can influence or correlate with emotional states, and

(3) offer a foundation for building emotion-aware lighting environments or therapeutic systems. Applications of this system can extend to smart home automation, clinical stress monitoring, emotion-responsive lighting in workspaces, or even mood-based music and light experiences.

This multidisciplinary project combines electronics, physiology, embedded systems, and data visualization, offering practical insight into human-computer interaction and affective computing. By combining hardware interfacing, data acquisition, and real-time visualization, the system serves as a compact and affordable tool for investigating the psycho-physiological effects of environmental light on human emotional states.

Keywords:

- Bio-signal Monitoring
- Human Emotion Detection
- Raspberry Pi Embedded System

2. INTRODUCTION:

Light is a powerful environmental factor that significantly influences human emotion, behaviour, and physiological functioning. Variations in light intensity and spectral composition affect circadian rhythms, sleep–wake cycles, hormonal regulation, and neural activity, thereby playing a crucial role in mood regulation, alertness, stress levels, and overall emotional well-being. With the increasing prevalence of artificial lighting in modern environments, understanding the interaction between light exposure and human emotional states has become increasingly important.

This project investigates the physiological effects of controlled light stimuli on emotional states through the analysis of bioelectrical signals using non-invasive electrophysiological sensing techniques. The system utilizes the BioAmp EXG Pill to acquire EEG, EMG, and ECG signals, an ADS1115 high-resolution analog-to-digital converter for accurate signal sampling, and a Raspberry Pi 4B as the central processing and control unit for real-time data acquisition, processing, and visualization. By systematically varying light intensity and colour temperature, the project aims to correlate changes in bio-signals with emotional responses, enabling objective interpretation of mood states.

The proposed system is intended for application in several real-world scenarios, including:

- Optimized lighting environments to enhance emotional well-being
- Adaptive lighting systems that dynamically adjust colour temperature and intensity based on user mood
- Stress-reducing and productivity-enhancing lighting in office spaces
- Light-based therapeutic interventions for Seasonal Affective Disorder (SAD) and depression
- Hospital lighting optimization to improve patient comfort and recovery outcomes
- Mood-responsive retail lighting to influence customer perception and behaviour
- Smart fitting-room lighting that adapts to enhance clothing appearance and user experience

The novelty of this project lies in its integration of real-time emotional state analysis with ambient light modulation using low-cost, compact hardware. Unlike conventional approaches that rely on subjective self-reporting or expensive medical-grade instrumentation, this system employs accessible, non-invasive electrophysiological monitoring combined with Raspberry

Pi-based edge processing to create a portable, scalable, and cost-effective emotional-response detection platform. Furthermore, by directly correlating specific light parameters with EEG, EMG, and ECG patterns, the project provides a practical framework for studying the neurophysiological basis of light-induced emotional modulation in real-world environments.

4. Working Principle:

a) Signal Acquisition

The BioAmp EXG Pill acquires non-invasive electrophysiological signals, including EEG, EMG, and ECG, from the subject while being exposed to controlled light stimuli of varying intensity and colour temperature. Electrodes are positioned according to standardized placements to ensure signal reliability. An initial electrode integrity and baseline stability check is performed to verify proper contact and eliminate floating or noisy inputs before data acquisition begins.

b) Signal Amplification and Analog-to-Digital Conversion

The low-amplitude bioelectrical signals are amplified and conditioned by the BioAmp EXG Pill to bring them within a measurable voltage range. These conditioned analog signals are then digitized using the ADS1115 16-bit analog-to-digital converter, which provides high-resolution sampling and improved noise performance. The ADS1115 communicates with the Raspberry Pi 4B via the I²C interface, enabling reliable and precise real-time data transfer.

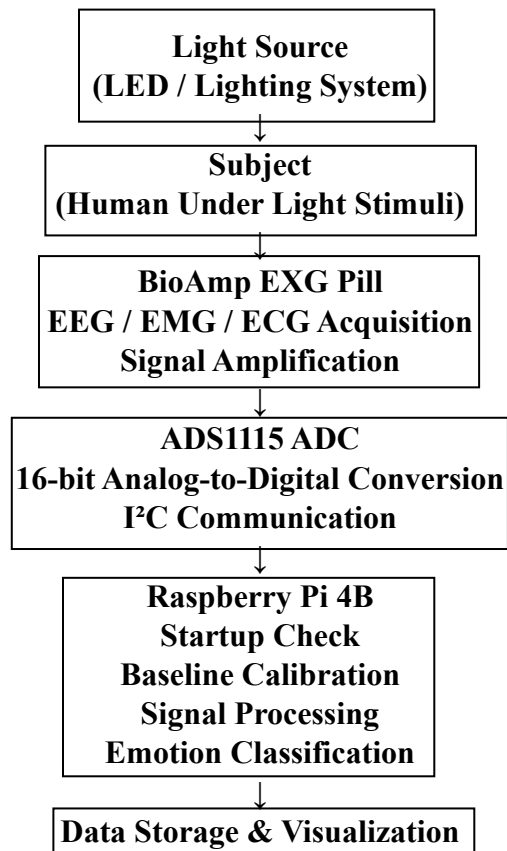
c) Signal Processing, Feature Extraction, and Emotional Analysis

The Raspberry Pi 4B performs real-time signal processing, including baseline correction, artifact rejection, and band-limited filtering to isolate physiologically relevant frequency components. Features such as EEG band power ratios (e.g., alpha, beta, and theta activity), EMG activity levels, and ECG-derived metrics are extracted and analysed. These physiological features are then correlated with controlled light parameters (intensity and colour temperature) to infer emotional states. The processed data and inferred emotional responses are logged and visualized for further analysis and interpretation.

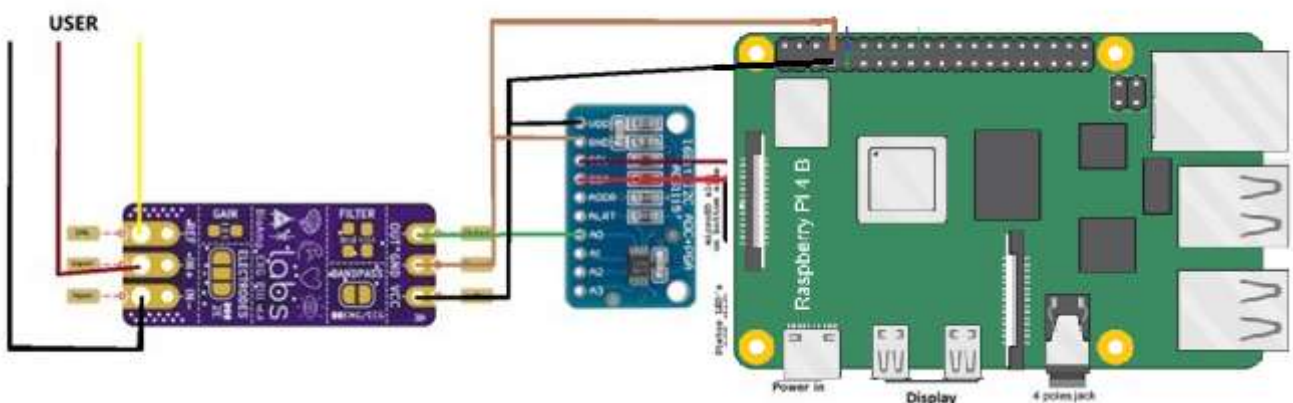
2025-12-16 14:54:02.32, 0.56 V	
2025-12-16 14:54:02.38, 0.57 V	
2025-12-16 14:54:02.45, 1.91 V	<----- Sharp impulse
2025-12-16 14:54:02.51, 0.58 V	
2025-12-16 14:54:02.57, 0.55 V	

2025-12-16 14:53:54.812, 0.777375	\	
2025-12-16 14:53:54.873, 1.031500		
2025-12-16 14:53:54.934, 1.338625		
2025-12-16 14:53:54.995, 1.598125		
2025-12-16 14:53:55.056, 1.771125		
2025-12-16 14:53:55.117, 1.885625		
2025-12-16 14:53:55.178, 1.951625		
2025-12-16 14:53:55.239, 1.924875	-----	Gradual Change
2025-12-16 14:53:55.300, 1.788250		
2025-12-16 14:53:55.361, 1.601250		
2025-12-16 14:53:55.422, 1.316875		
2025-12-16 14:53:55.483, 1.082875		
2025-12-16 14:53:55.544, 0.859375		
2025-12-16 14:53:55.605, 0.705375	/	

5. Flowchart:



6. Circuit Diagram:



BioAmp EXG Output	ADC A0	
ADC SCL	Pin 5 - GPIO3 (SCL)	
ADC SDA	Pin 3 - GPIO2 (SDA)	
BioAmp EXG VCC	ADC VDD	Pin1/17 - +3.3V
BioAmp EXG GND	ADC GND	Pin 6 - GND

Fig 2: Circuit Diagram of our Prototype

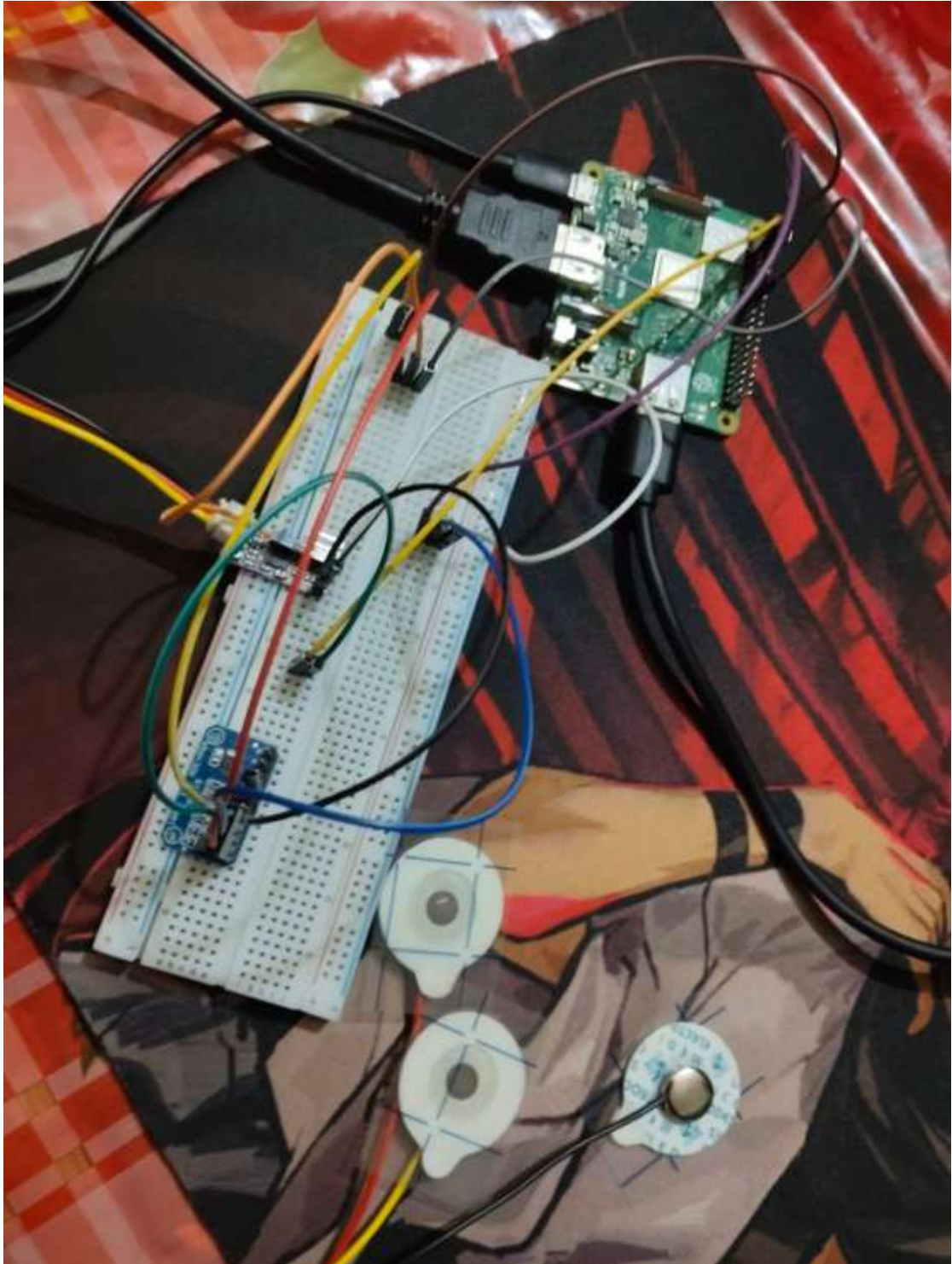


Fig 3: Our Initial Prototype

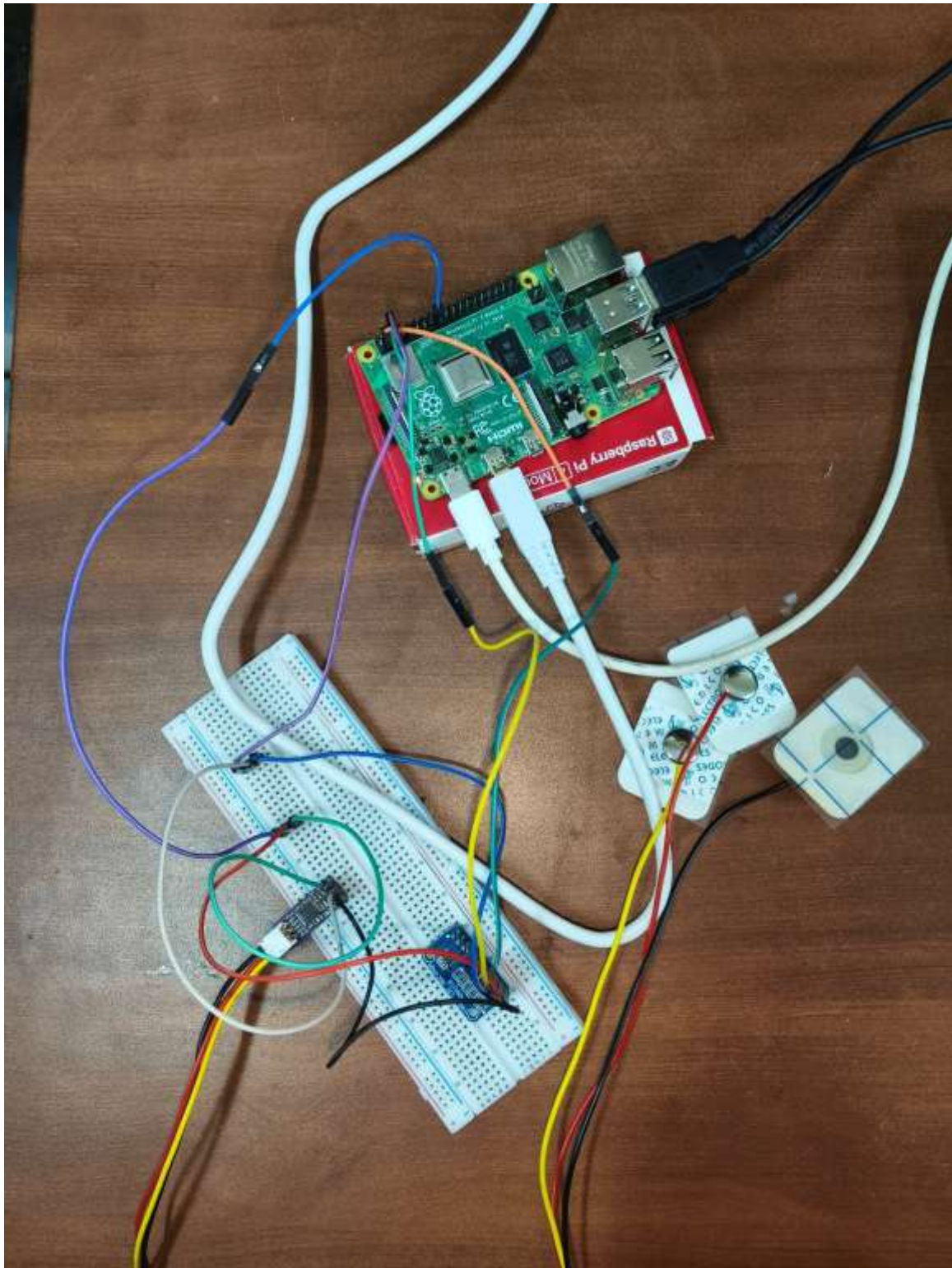


Fig 3: Our Current Model

7. Components:

COMPONENT	DESCRIPTION
BioAmp EXG Pill	A compact, open-source analog front-end module designed for non-invasive electrophysiological signal acquisition , including EEG, EMG, and ECG. It provides high input impedance, low-noise amplification, and basic signal conditioning, enabling reliable capture of microvolt-level bio-signals from the human body.
ADS1115 (16-bit ADC)	A high-resolution 16-bit analog-to-digital converter with four input channels and programmable gain amplifier (PGA). It converts the amplified analog bio-signals into digital form with high precision and communicates with the Raspberry Pi via the I²C interface , ensuring accurate real-time data acquisition.
Raspberry Pi 4B	A powerful single-board computer running a Linux-based operating system, used as the central processing unit of the system. It handles I ² C communication with the ADS1115, performs real-time signal processing, baseline calibration, noise filtering, feature extraction , emotion classification, and controls light stimuli via GPIO/PWM interfaces.
Electrodes (Gel / Ag–AgCl)	Non-invasive surface electrodes placed on the skin to acquire bioelectrical signals. Typically made of silver/silver-chloride (Ag–AgCl) , they provide low electrode–skin impedance and stable signal quality, essential for accurate EEG/EMG/ECG measurements.
Light Source (LED / RGB Lighting Module)	A controllable white or RGB LED lighting system used to apply varying light intensity and colour temperature as experimental stimuli. The light source is interfaced with the Raspberry Pi to enable adaptive and dynamic lighting control based on detected emotional states.
Power Supply (5V Regulated)	A stable 5V power source used to supply the Raspberry Pi 4B and associated peripherals. Proper regulation ensures consistent ADC performance and minimizes noise introduced into sensitive bio-signal measurements.

1. BioAmp EXG Pill

1. Technical Overview:

The BioAmp EXG Pill is an open-source, low-noise analog front-end designed to acquire extremely low-amplitude electrophysiological signals produced by the human body. These signals typically lie in the microvolt range and require careful amplification and conditioning before digital processing.

2. Key Specifications:

- Supports EEG, EMG, and ECG signal acquisition
- High input impedance to prevent signal attenuation
- Differential input configuration (IN+ and IN-)
- Dedicated reference electrode (REF)
- Low-noise instrumentation amplifier
- Operates at 3.3 V supply
- Compact and portable design

3. Functionality:

The BioAmp EXG Pill performs the initial and most critical stage of signal acquisition. Surface electrodes detect bioelectrical activity, which is then differentially amplified to reject common-mode noise such as power-line interference. The conditioned analog output is suitable for precise analog-to-digital conversion.

4. Usability:

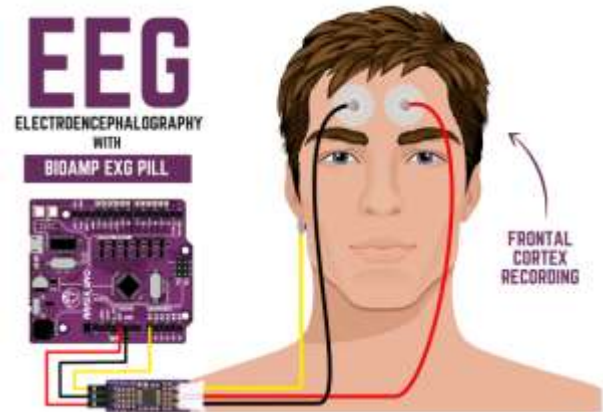
The device is completely non-invasive and safe for human use. It is easy to integrate with external ADCs such as the ADS1115 and requires minimal setup, making it ideal for academic and experimental applications.

5. Potential Flexibility:

The BioAmp EXG Pill can be configured for EEG, EMG, or ECG by changing electrode placement. Multiple BioAmp units can be combined for multi-channel acquisition, and it is compatible with various microcontrollers and single-board computers.

6. Importance in the Project:

The BioAmp EXG Pill ensures accurate and reliable bio-signal capture, which is essential for meaningful emotion analysis under controlled lighting conditions.



2. ADS1115 (16-bit Analog-to-Digital Converter)

1. Specifications:

- 16-bit resolution
- Four analog input channels
- Programmable Gain Amplifier (PGA)
- I2C communication interface
- Sampling rate up to 860 samples per second

2. Functionality:

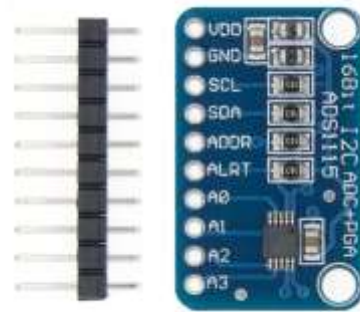
The ADS1115 converts amplified analog bio-signals into high-resolution digital data for processing by the Raspberry Pi.

3. Usability:

Its high precision, simple wiring, and extensive software support make it reliable for continuous biomedical signal acquisition.

4. Potential Flexibility:

Multiple channels allow sensor expansion, and adjustable gain settings enable



adaptation to different signal amplitudes.

3. Raspberry Pi 4B

1. Technical Overview:

The Raspberry Pi 4B is a high-performance single-board computer that serves as the central processing and control unit of the system. It runs a Linux-based operating system and supports advanced signal processing and real-time control.

2. Key Specifications:

- Quad-core ARM Cortex-A72 processor
- Clock speed up to 1.5 GHz
- Up to 4 GB RAM
- Built-in I2C, SPI, UART, GPIO, and PWM interfaces
- HDMI and USB connectivity
- Runs Raspberry Pi OS (Linux)

3. Functionality:

The Raspberry Pi 4B receives digitized bio-signal data from the ADS1115 via I2C communication. It performs baseline calibration, noise filtering, feature extraction, and emotion classification. Based on the detected emotional state, it controls the lighting system using GPIO or PWM outputs. It also handles data storage and visualization.

4. Usability:

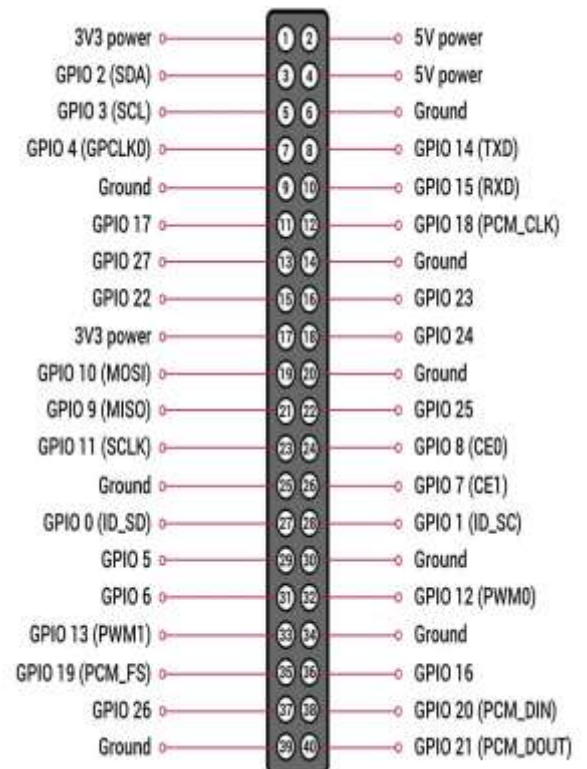
The Raspberry Pi 4B provides a robust software ecosystem with extensive Python libraries for signal processing and visualization. Its ease of debugging and development makes it suitable for real-time biomedical experiments.

5. Potential Flexibility:

The platform supports machine learning integration, cloud connectivity, IoT expansion, and scalable system architectures. It can be extended to multi-sensor setups or portable emotion-aware systems.

6. Importance in the Project:

The Raspberry Pi 4B enables real-time analysis and adaptive lighting control, transforming raw bio-signals into meaningful emotional insights.



4. Electrodes (Ag-AgCl / Gel Electrodes)

1. Specifications:

- Silver/Silver-Chloride material
- Non-invasive surface electrodes
- Low skin-electrode impedance

2. Functionality:

Electrodes serve as the interface between the human body and the BioAmp, enabling accurate collection of bioelectrical signals.

3. Usability:

They are comfortable for users, easy to place, and provide stable signal quality for long-duration recordings.

4. Potential Flexibility:

Electrodes can be repositioned for different physiological signals and expanded for multi-channel recording.



5. Light Source (White / RGB LED)

1. Specifications:

- Adjustable brightness and colour temperature
- PWM controllable
- Low power consumption

2. Functionality:

The light source provides controlled visual stimuli to evaluate physiological and emotional responses under varying lighting conditions.

3. Usability:

It is safe, energy-efficient, and easy to control using Raspberry Pi GPIO pins.

4. Potential Flexibility:

The system can be extended to smart lighting solutions, adaptive environments, and mood-based lighting applications.

6. Power Supply (5 V Regulated)

1. Specifications:

- Regulated 5 V DC output
- Stable current delivery
- Low noise output

2. Functionality:

Provides consistent power to the Raspberry Pi and connected peripherals.

3. Usability:

Standard power supplies are widely available and reliable for continuous operation.

4. Potential Flexibility:

Can be replaced with battery-based or portable power solutions.

7. Wires and Breadboard / PCB

1. Specifications:

- Jumper wires
- Solderless breadboard or custom PCB

2. Functionality:

Provides electrical interconnections between system components during development and deployment.

3. Usability:

Allows rapid prototyping, easy troubleshooting, and low-cost experimentation.

4. Potential Flexibility:

Can transition from breadboard to PCB for compact, robust, and noise-resistant system designs.

8. Results and Inference:

i. Experimental Test Case Description

The experimental evaluation was conducted to analyze how different lighting conditions influence emotional states, as inferred from bioelectrical signals acquired using the BioAmp EXG Pill, ADS1115 ADC, and Raspberry Pi 4B. Four participants were exposed to five controlled conditions: **Natural Sunlight** (control), a **Happy/Funny video**, a **Warm/Mild video**, a **Light flicker** and a **Sudden Image**.

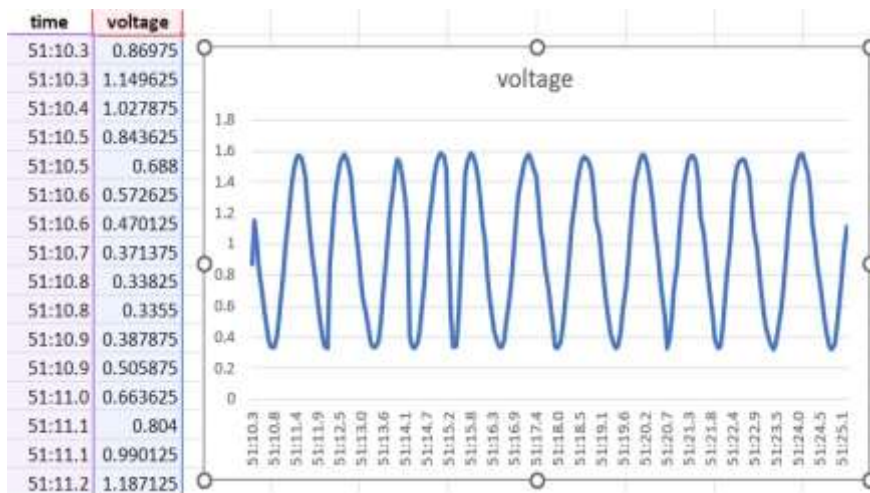
For each condition, EEG signals were recorded into **.csv files** and mapped. The obtained percentages represent the relative dominance of each emotional state during exposure to the specific condition.

ii. Participant 1 – Scenario Description and Interpretation

Participant 1 is exposed to Visual Stimuli to check for moderate emotional variability and distinct responses to changes in conditions.

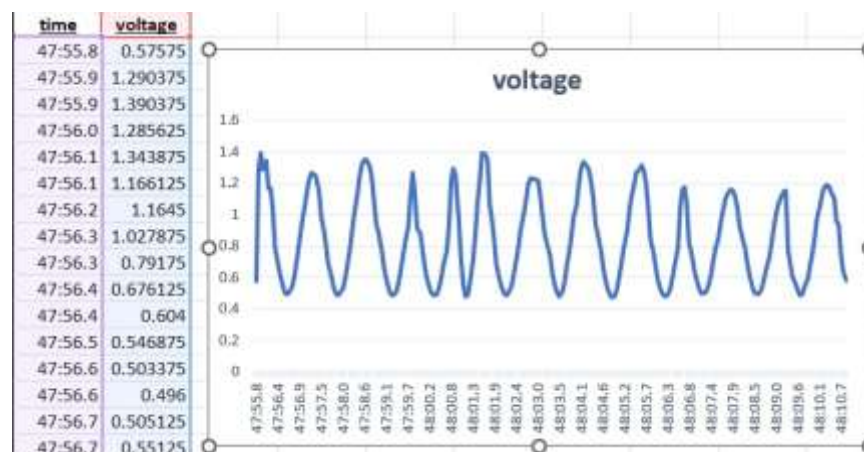
1.1. Natural Light (Control Condition):

Participant 1 readings were recorded while looking at natural sunlight.



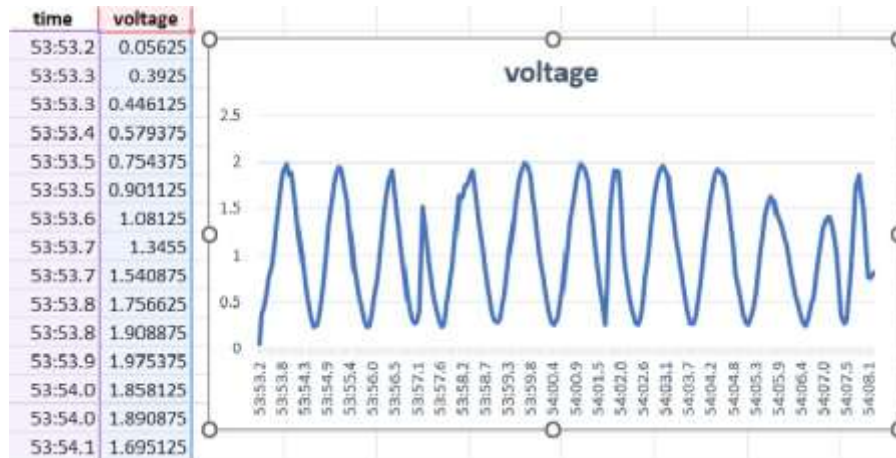
1.2. Exposure to Funny/Happy Video:

Participant 1 readings were recorded while looking at a funny/happy video.



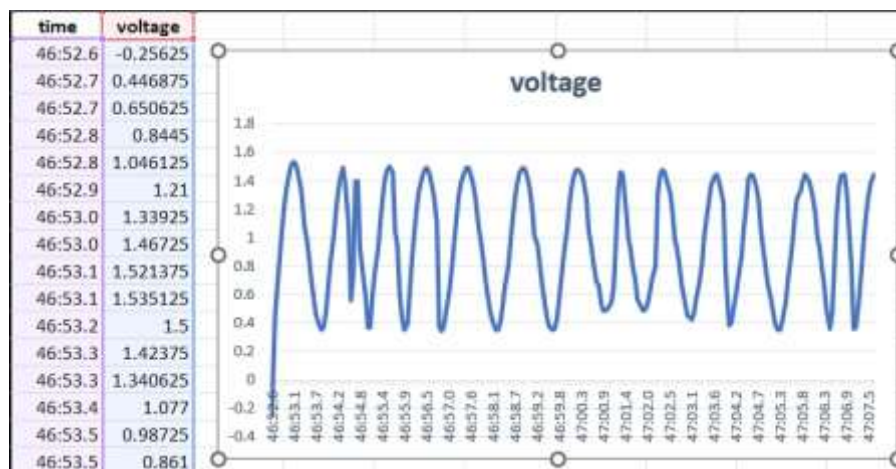
1.3. Exposure to Warm/Mild Video:

Participant 1 readings were recorded while looking at a Warm/Mild video.



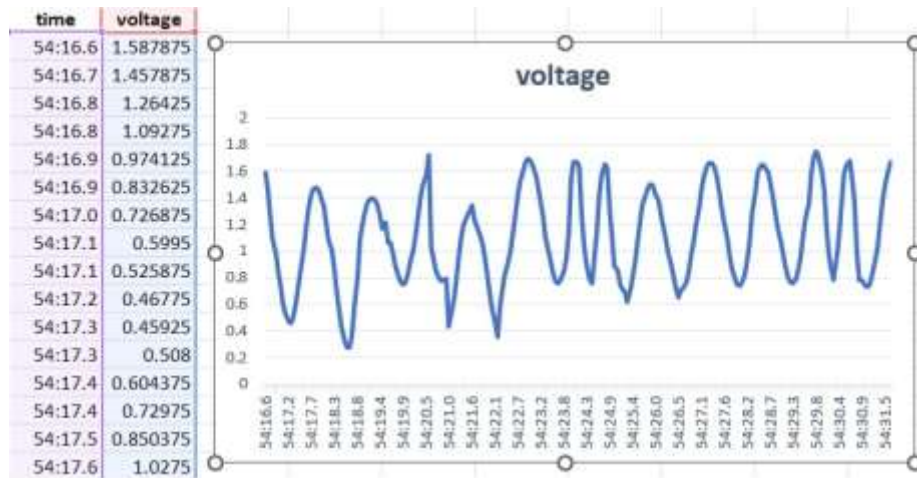
1.4. Exposure to Sudden image:

Participant 1 readings were recorded when shown a sudden image.



1.5. Exposure to Light Flicker:

Participant 1 readings were recorded when subjected to flickering light.



Inference:

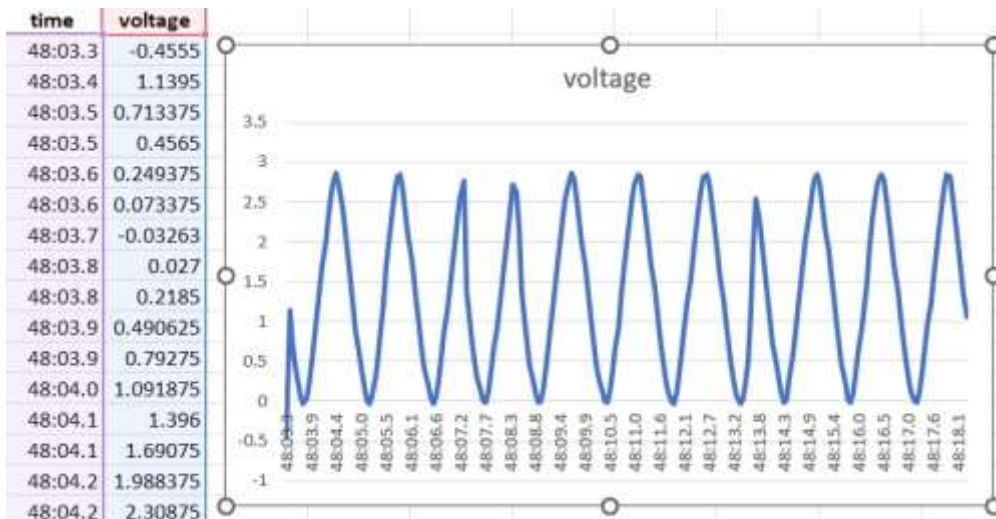
Thus, we see that the Participant 1 has a neutral reaction to long form video content and has a sharp reaction to sudden or unexpected conditions.

iii. Participant 2– Scenario Description and Interpretation

Participant 2 is exposed to Visual Stimuli to check for moderate emotional variability and distinct responses to changes in conditions.

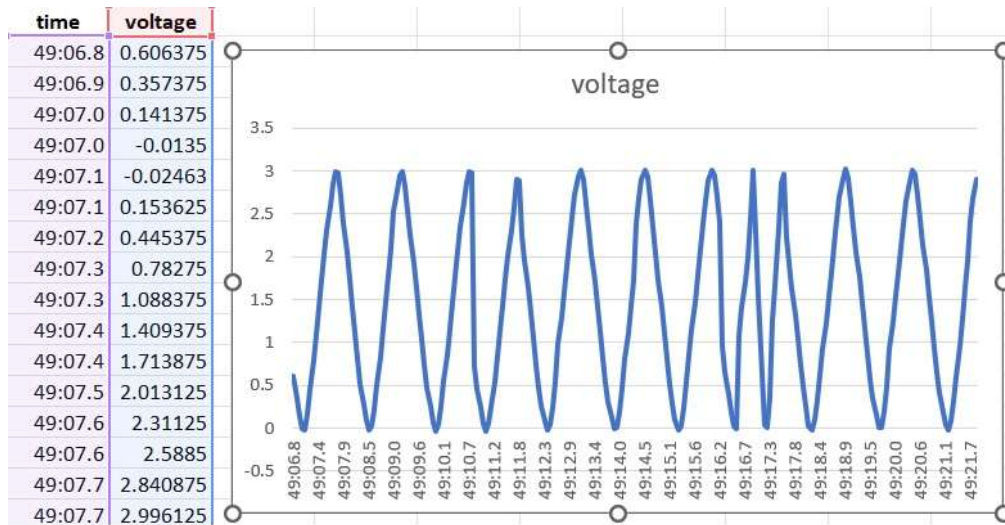
2.1.Natural Light (Control Condition):

Participant 2 readings were recorded while looking at natural sunlight.



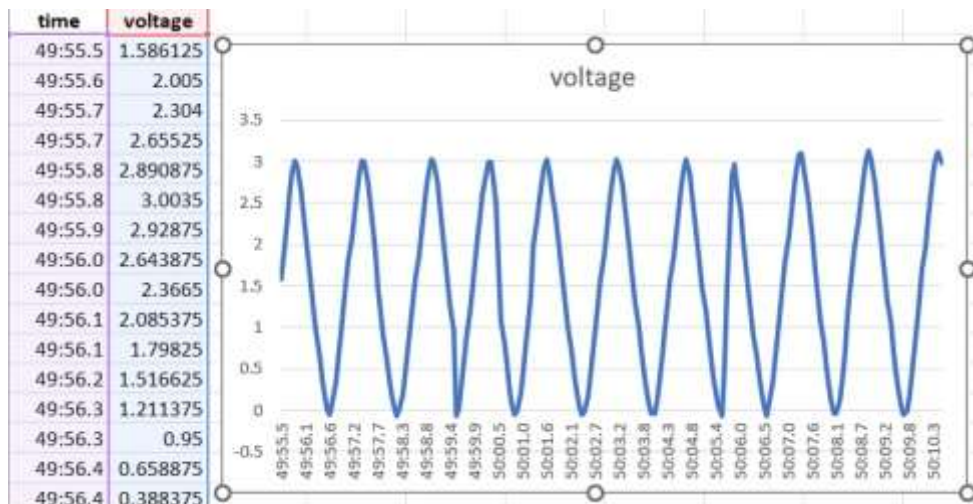
2.2.Exposure to Funny/Happy Video:

Participant 2 readings were recorded while looking at funny/ happy video.



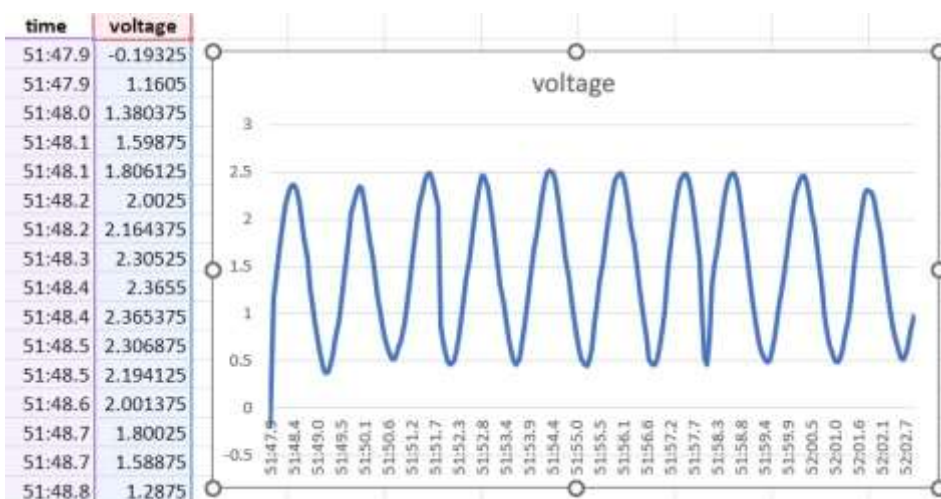
2.3.Exposure to Warm/Mild Video:

Participant 2 readings were recorded while looking at warm/mild video.



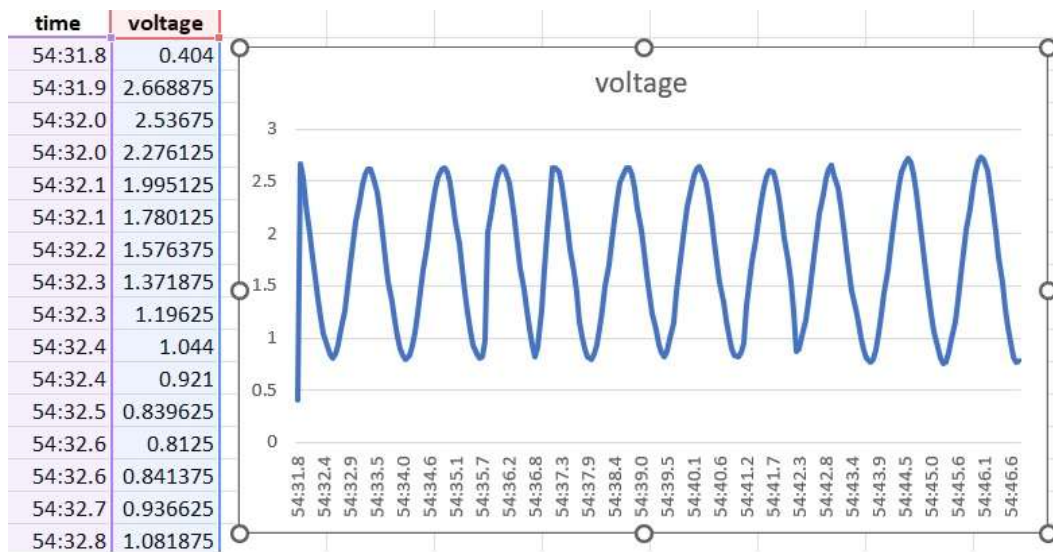
2.4.Exposure to Sudden image:

Participant 2 readings were recorded when shown a sudden image.



2.5.Exposure to Light Flicker:

Participant 2 readings were recorded when subjected to flickering light.



Inference:

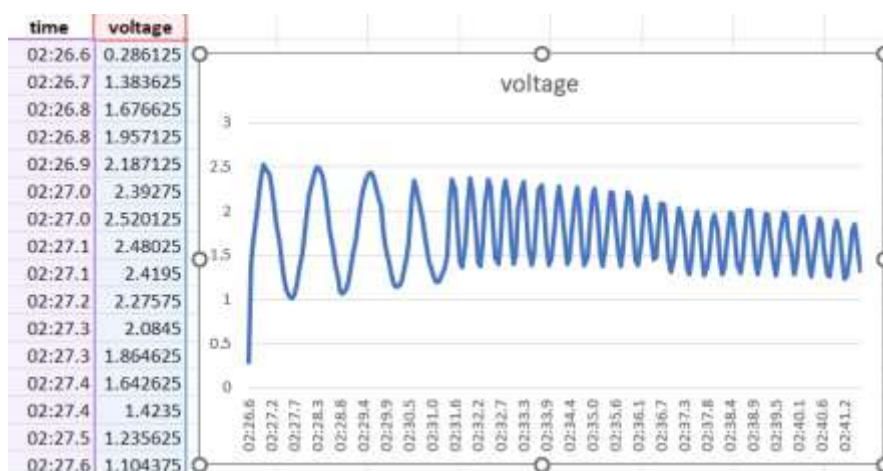
Thus, we see that the Participant 2 has a neutral reaction to most long form video content and has a sharp reaction to sudden or unexpected image or a flash.

iv. Participant 3– Scenario Description and Interpretation

Participant 3 is exposed to Visual Stimuli to check for moderate emotional variability and distinct responses to changes in conditions.

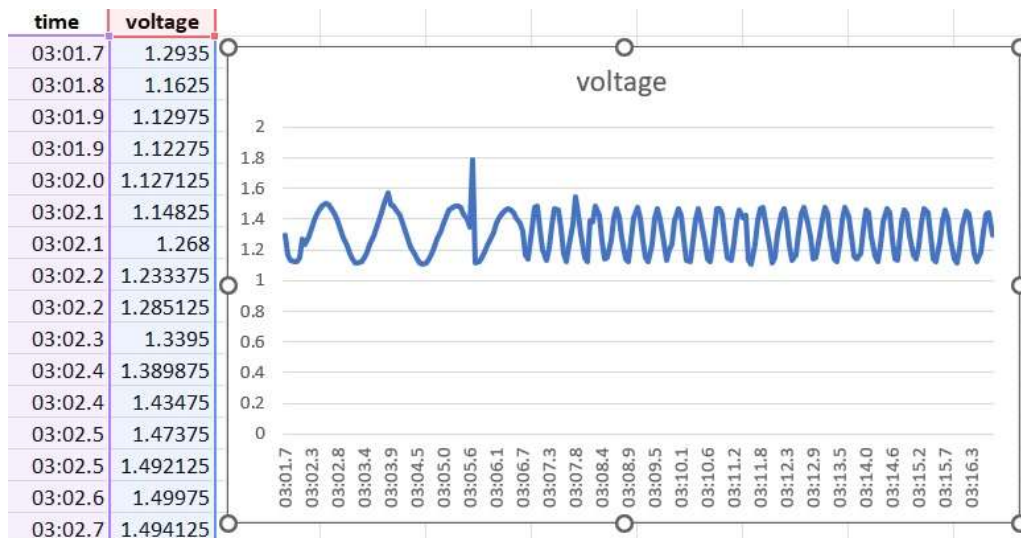
3.1.Natural Light (Control Condition):

Participant 3 readings were recorded while looking at natural sunlight.



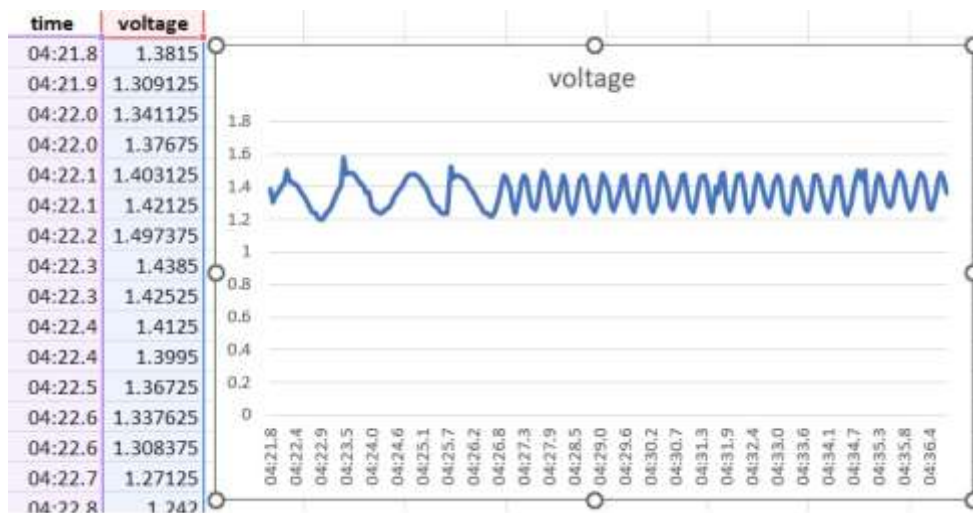
3.2.Exposure to Funny/Happy Video:

Participant 3 readings were recorded while looking at funny/ happy video.



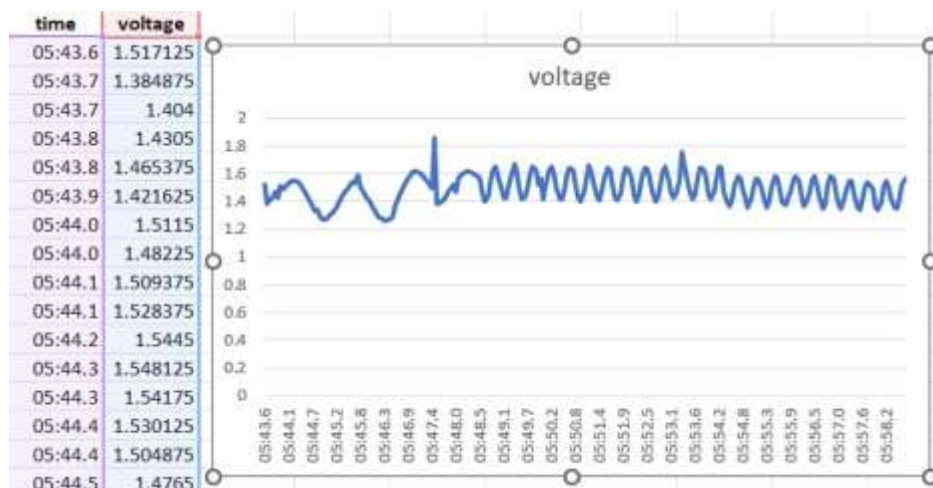
3.3.Exposure to Warm/Mild Video:

Participant 3 readings were recorded while looking at warm/mild video.



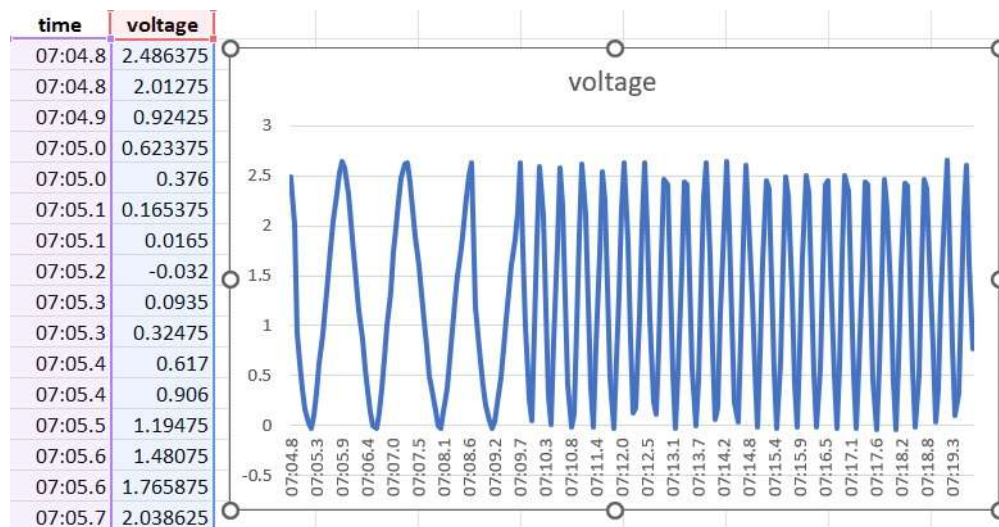
3.4.Exposure to Sudden image:

Participant 3 readings were recorded when shown a sudden image.



3.5.Exposure to Light Flicker:

Participant 3 readings were recorded when subjected to flickering light.



Inference:

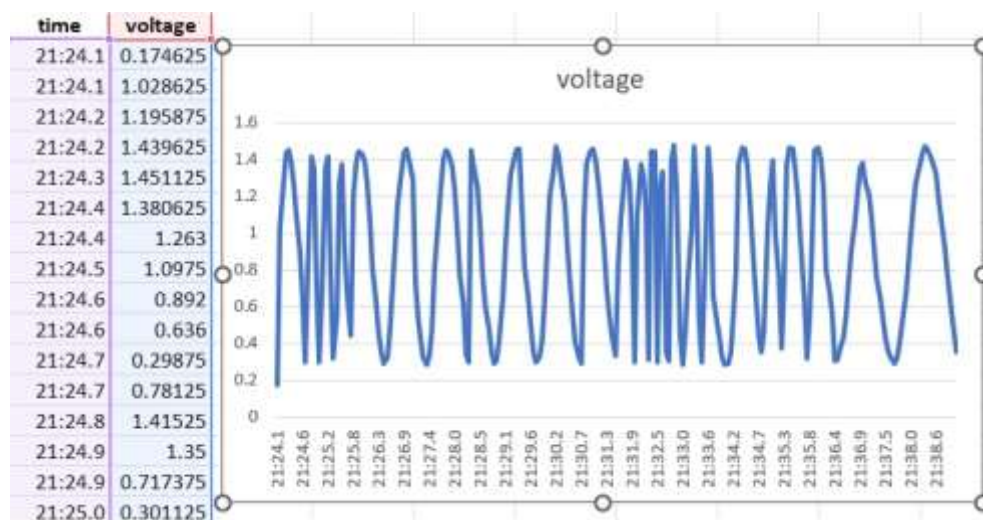
Thus, we see that the Participant 3 has an extreme reaction to most long form video content and to sudden or unexpected image or a flash with heightened activity afterwards.

v. Participant 4– Scenario Description and Interpretation

Participant 4 is exposed to Visual Stimuli to check for moderate emotional variability and distinct responses to changes in conditions.

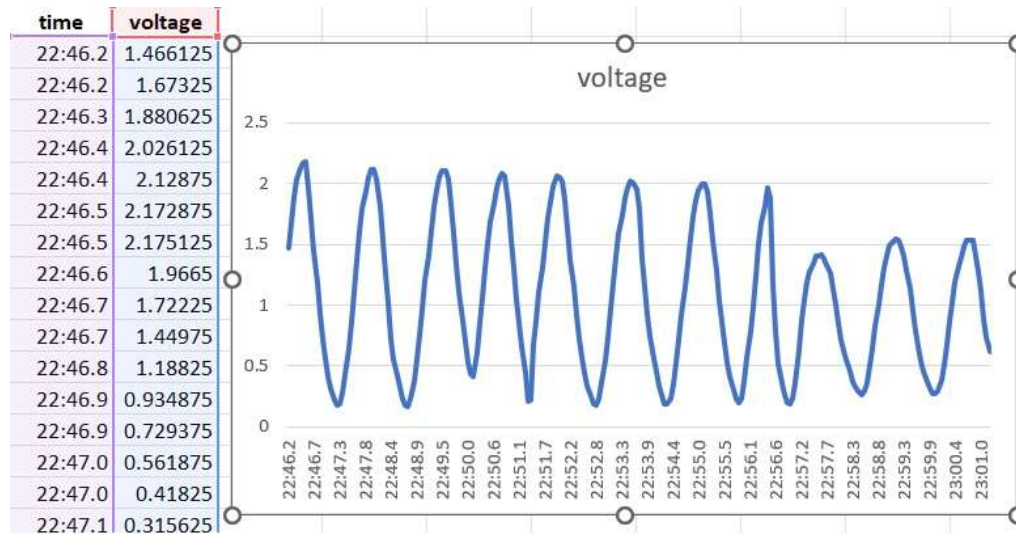
4.1.Natural Light (Control Condition):

Participant 4 readings were recorded while looking at natural sunlight.



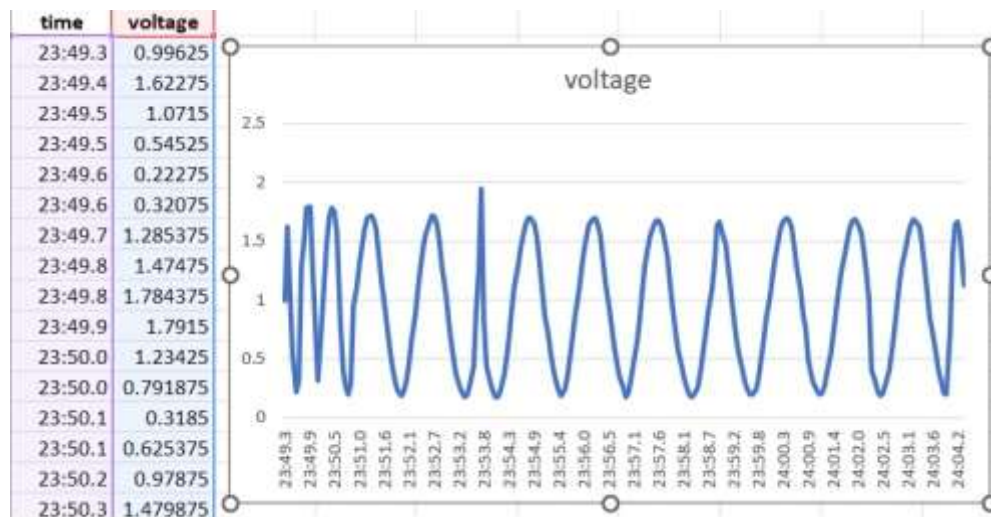
4.2.Exposure to Funny/Happy Video:

Participant 4 readings were recorded while looking at funny/ happy video.



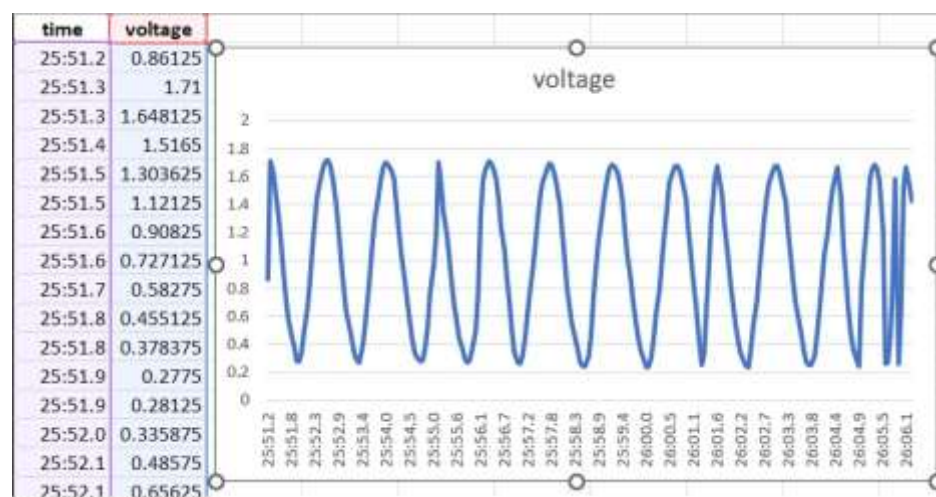
4.3.Exposure to Warm/Mild Video:

Participant 4 readings were recorded while looking at warm/mild video.



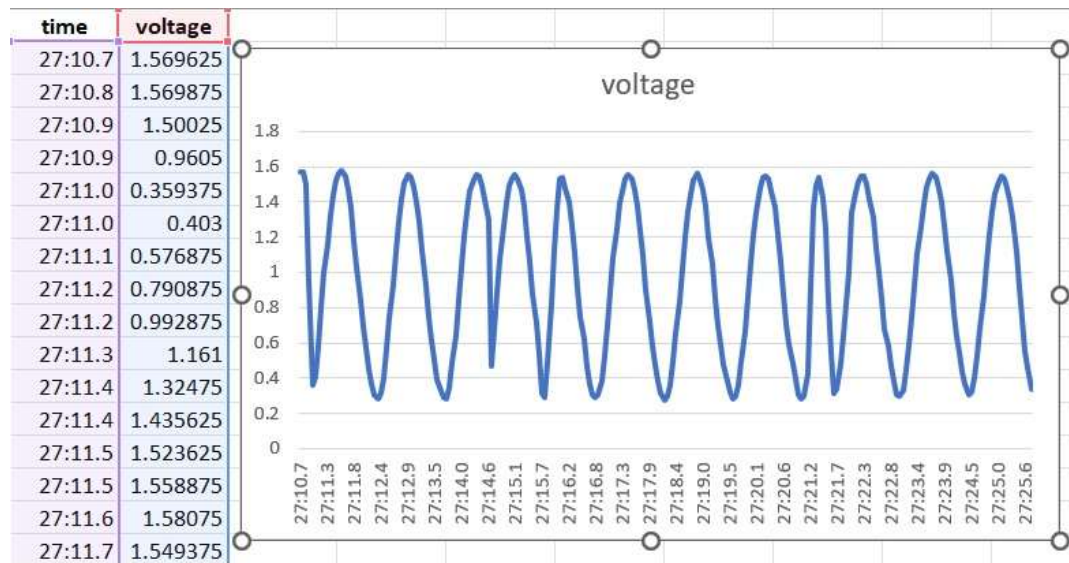
4.4.Exposure to Sudden image:

Participant 4 readings were recorded when shown a sudden image.



4.5.Exposure to Light Flicker:

Participant 4 readings were recorded when subjected to flickering light.



Inference:

Thus, we see that the Participant 4 has an moderate reaction to most long form video content and to sudden or unexpected image or a flash and extreme reaction to sad/monotonous videos.

vi. Overall Inference

The experimental results confirm that visual stimuli influence emotional states and that these effects can be reliably detected using non-invasive bio-signal analysis. Variations between participants emphasize the need for adaptive and personalized emotion-aware lighting systems. The results validate the effectiveness of the proposed system for applications in smart environments, mental well-being, and human-centric design.

9. Conclusion:

This project presents a robust and low-cost framework for investigating the influence of light on human emotional states through real-time analysis of electrophysiological bio-signals. By integrating the BioAmp EXG Pill, 16-bit ADS1115 analog-to-digital converter, and the Raspberry Pi 4B, the system successfully acquires, processes, and interprets non-invasive EEG signals under controlled lighting conditions. The use of high-resolution signal conversion and real-time processing enables accurate correlation between light intensity and color variations and the corresponding physiological responses.

The proposed system demonstrates a portable, safe, and scalable solution for emotion monitoring, making it suitable for applications beyond laboratory environments. Its modular architecture supports future enhancements such as adaptive lighting control, machine learning-based emotion classification, and cloud-based data analysis. The findings of this work highlight the potential of accessible embedded systems in bridging the gap between neuroscience, environmental interaction, and human-centric design. Overall, this project lays a strong foundation for the development of emotion-aware technologies that can contribute to improved mental well-being, smart environments, and personalized user experiences.

10. Future Scope of Application:

1. Smart and Adaptive Lighting Systems: The proposed system can be extended into fully automated smart lighting environments by integrating the Raspberry Pi 4B with home automation platforms. Real-time emotion detection from EEG signals can be used to dynamically adjust light intensity and colour temperature to suit the user's emotional state. Such adaptive lighting systems can enhance comfort, relaxation, and alertness in residential and commercial spaces, promoting overall emotional well-being.

2. Continuous Mental Health Monitoring: With further refinement, the system can be adapted for long-term, passive emotional monitoring. By continuously tracking physiological responses to light stimuli, early indicators of emotional stress, anxiety, or mood imbalance can be identified. This non-invasive approach offers a potential tool for supporting mental health awareness without the need for constant user interaction or subjective self-reporting.

3. Workplace and Productivity Optimization: The system can be deployed in workplace environments to create emotion-aware lighting conditions. By detecting stress or fatigue through bio-signal analysis, lighting parameters can be adjusted in real time to maintain alertness and reduce cognitive load. Such intelligent lighting systems can contribute to increased productivity, reduced stress levels, and improved employee well-being.

4. Virtual Reality (VR) and Augmented Reality (AR) Integration: The emotion detection framework can be integrated with VR and AR platforms to enhance immersive user experiences. Real-time emotional feedback can be used to modify virtual lighting, environment tone, or scene dynamics, creating more responsive and emotionally adaptive virtual environments for gaming, training, and simulation applications.

5. Clinical and Therapeutic Applications: The system holds significant potential for clinical research and therapeutic use, particularly in studying light-based interventions for conditions such as Seasonal Affective Disorder (SAD) and depression. Controlled exposure to specific lighting conditions, combined with real-time physiological feedback, can help researchers evaluate the effectiveness of therapeutic light treatments and optimize personalized therapy protocols.

6. Emotion-Aware Smart Environments: Beyond lighting, the system can be expanded to control additional environmental parameters such as sound, temperature, and visual ambience. This enables the development of emotion-aware smart environments that respond holistically to human emotional states, improving comfort in homes, hospitals, and public spaces.

7. Machine Learning–Based Emotion Classification: Future versions of the system can incorporate machine learning and artificial intelligence algorithms to improve emotion classification accuracy. By training models on larger datasets collected via the Raspberry Pi 4B, more nuanced emotional patterns can be identified, enabling personalized and adaptive responses over time.

8. Scalable and IoT-Enabled Deployments: With network connectivity, the system can be extended into IoT-based architectures, allowing remote monitoring, cloud data storage, and large-scale deployment. This scalability makes the solution suitable for smart cities, healthcare monitoring systems, and research studies involving multiple users.

11. References:

- i. Fernández, D. C., Fogerson, P. M., Lazzerini Ospri, L., Thomsen, M. B., Layne, R. M., Severin, D., Zhan, J., Singer, J. H., & Hattar, S. (2018). **Light affects mood and learning through distinct retina–brain pathways.** *Cell*, 175(1), 71–84.
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